HUY DUC (JUSTIN) VU

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SUMMARY

AI-focused Data Science graduate candidate with hands-on experience building production-ready end-to-end ML pipelines, RAG, and cloud data architectures that support measurable business impact by translating raw data into business intelligent, and scalable solutions. Currently finalising a Master degree in Data Science at UTS with a GPA of 6.7/7.

EDUCATION

University of Technology Sydney **Feb 2025 – Dec 2026 (Expected)**

Master of Data Science and Innovation *GPA: 6.7/7*

University of Wollongong **Feb 2022 – Dec 2024**

Bachelor of Business, Business Analytics and Finance *WAM: 86/100*

- **Dean's Merit List** 2022 & 2023; Ranked **Top 3 in 8 subjects** across Business Analytics, Database System, Statistics, and Finance.

SKILLS

- **Programming languages**: Python (OOP, Scikit-learn, PyTorch, TensorFlow, spaCy, NLTK, NumPy), SQL, R.
- **AI/ML**: MLOps, NLP, LLM, VLM, RAG, LangChain, Ollama, Claude Code.
- **Cloud & Databases**: GCP, Azure, Snowflake, Databricks, Airflow, dbt, BigQuery; MySQL, PostgreSQL, MongoDB.
- **Visualisation & Tools**: Power BI, Tableau, SAS Viya, Excel, SPSS Statistics, Git, Docker.

CERTIFICATIONS

- Microsoft Certified: **Power BI Data Analyst Associate** (PL-300) **Oct 2026**

WORK EXPERIENCE

Vennu Australia | Business Data Analyst (Casual) **Sydney, NSW | Feb 2024 – Nov 2025**

Vennu is a marketplace platform that lists spaces and connects with guests looking to book local, affordable venues for events.

- Architected end-to-end **ELT pipelines** (Python, Google Maps API) integrating **10+ data sources** into a unified **Tableau dashboard**, boosting strategic decision-making **efficiency by 135%** across **5+ stakeholder** teams.
- Drove a **data-led marketing campaign** by analysing customer engagement patterns, increasing **social media reach by 120%** and cutting the **business planning** cycle from 2 weeks to **under 3 days (over 70% reduction)**.
- Translated ambiguous stakeholder requirements into **scalable data projects**, iterating on prototypes through structured feedback loops to **deliver production-ready analytics solutions**.

Vennu Australia | Business Analyst Intern **Sydney, NSW | Nov 2023 – Feb 2024**

- Engineered **4 Power BI dashboards** by extracting and transforming data from **6 MySQL sources**, reducing reporting turnaround from 1–2 weeks to **2–3 days**, a **70% time saving**.
- Produced a **board-level** proposal to adopt **Power BI** in daily operations, defining short- and long-term objectives; the initiative was approved and later scoped to include **ML pipeline** integration.
- Delivered 2023 annual business review **using Excel** and **Power BI** (including writing **DAX**), providing actionable insights adopted into the company's operational roadmap.

UOW Chinese Academic Business Learning & Education (CABLE) | Senior Leader **Wollongong, NSW | Sep 2023 – Dec 2024**

- Designed and facilitated **Introduction to Statistics** workshops for **50+ participants** covering regression, probability distributions, and hypothesis testing, achieving a **99% satisfaction rate** in post-session feedback.

PROJECTS

Multimodal RAG for Intelligent CCTV Querying | RAG, LangChain, ChromaDB, Gemini VLM, Gemini Embedding 2, Ollama

- Architected 3-phase **multimodal RAG** pipeline: ingesting raw footage via FFmpeg **temporal chunking**, **embeddings into ChromaDB**, synthesising answers with **Gemini VLM**, enabling **natural-language querying** across CCTV footage replacing manual review.
- Validated retrieval quality across **264 test cases** on a **128-hour dataset**, achieving **73% pass rate**, with **86% event retrieval accuracy** and **76% temporal correctness**, evaluated using an automated **LLM-as-a-Judge** framework.

ELT Data Pipelines with Airbnb & Census Data | Airflow, dbt, Postgres, GCP

- Engineered an automated **Airflow ELT pipeline** ingesting **300k+ Airbnb records** and ABS Census data into a **PostgreSQL data warehouse**, enabling large-scale analytics across 128 NSW LGAs and 200+ suburbs.
- Designed a **dbt medallion architecture** (Bronze / Silver / Gold) to power revenue prediction models, host segmentation, and demographic profiling, identifying key drivers of listing performance and host financial viability.